

Capacitance and Dielectrics, solved

Step-by-step solutions to every practice problem in Chapter 16. Try each problem on your own before you read its solution.

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P16.1

- **Capacitance: unchanged.** It depends only on geometry and the dielectric.
- **Voltage: doubles,** since $V = Q/C$.
- **Energy: quadruples,** since $U = Q^2/2C$.

P16.2

The field between the plates polarises the dielectric. Polar molecules partly line up with the field, and non-polar molecules become slightly stretched dipoles. Within the material, neighbouring dipoles cancel. At the two faces next to the plates, however, they leave thin layers of *bound* charge, negative next to the positive plate and positive next to the negative plate.

These bound charges produce a field that opposes the plates' field, reducing it to E_0/κ . For the same charge on the plates, the potential difference $V = Ed$ is therefore smaller, so $C = Q/V$ is larger. Equivalently, at a fixed voltage the plates can hold more charge, because the bound charges partly neutralise them.

P16.3

Charges: equal. In series, the same charge Q sits on each capacitor, because the inner plates and the wire joining them form an isolated, neutral conductor.

Voltages: $V = Q/C$, so the 1 μF capacitor has **10 times** the voltage of the 10 μF capacitor. The smaller capacitor takes most of the battery voltage, about 91% of it.

P16.4

The capacitor is isolated, so Q is constant.

- **Field:** $E = Q/(\epsilon_0 A)$ does not depend on d , so it is **unchanged**.
- **Voltage:** $V = Ed$ increases in proportion to d .
- **Energy:** $C = \epsilon_0 A/d$ decreases, so $U = Q^2/2C$ increases in proportion to d .

Source of the energy. The plates attract each other, with a force $\frac{1}{2}QE$ (Problem P16.15). Pulling them apart takes work, and that work becomes the extra field energy. The field now fills a larger volume at the same energy density $\frac{1}{2}\epsilon_0 E^2$.

P16.5

Capacitance:

$$C = \frac{\kappa\epsilon_0 A}{d} = \frac{5.4(8.854 \times 10^{-12})(0.050)^2}{2.0 \times 10^{-4}} = 6.0 \times 10^{-10} \text{ F} = 600 \text{ pF}.$$

Maximum voltage:

$$V_{\text{max}} = E_{\text{max}}d = (1.0 \times 10^8)(2.0 \times 10^{-4}) = 2.0 \times 10^4 \text{ V} = 20 \text{ kV}.$$

In practice, a safety factor of 2–5 would be applied.

P16.6

There are four distinct arrangements:

ARRANGEMENT	EQUIVALENT CAPACITANCE
All three in parallel	18 μF
All three in series	$6/3 = 2.0 \mu\text{F}$
Two in series, then in parallel with the third	$3 + 6 = 9.0 \mu\text{F}$

ARRANGEMENT	EQUIVALENT CAPACITANCE
Two in parallel, then in series with the third	$\frac{12 \times 6}{12 + 6} = 4.0 \mu\text{F}$

P16.7

Equivalent capacitance.

$$C_{23} = 2.0 + 6.0 = 8.0 \mu\text{F}, \quad C_{\text{eq}} = \frac{4.0 \times 8.0}{4.0 + 8.0} = 2.67 \mu\text{F}.$$

Total charge: $Q = C_{\text{eq}}V = 2.67 \times 24 = 64 \mu\text{C}$. This passes through C_1 and the parallel pair, which are in series.

- $V_1 = 64/4.0 = 16 \text{ V}$.
- $V_{23} = 64/8.0 = 8.0 \text{ V}$. The two add to 24 V. ✓
- $Q_2 = 2.0 \times 8.0 = 16 \mu\text{C}$ and $Q_3 = 6.0 \times 8.0 = 48 \mu\text{C}$. These add to 64 μC . ✓

P16.8

$$U = \frac{1}{2}CV^2 = \frac{1}{2}(1000 \times 10^{-6})(400)^2 = 80 \text{ J}, \quad P = \frac{80 \text{ J}}{1.0 \times 10^{-3} \text{ s}} = 80 \text{ kW}.$$

A small capacitor briefly delivers the power of a car engine.

P16.9

(a) Concentric spheres:

$$C = 4\pi\epsilon_0 \frac{ab}{b-a} = \frac{1}{8.99 \times 10^9} \cdot \frac{(0.10)(0.12)}{0.02} = 6.7 \times 10^{-11} \text{ F} = 67 \text{ pF}.$$

(b) The Earth:

$$C = 4\pi\epsilon_0 R = \frac{6.37 \times 10^6}{8.99 \times 10^9} = 7.1 \times 10^{-4} \text{ F} = 710 \mu\text{F}.$$

The whole planet has less capacitance than a single large electrolytic capacitor. This shows again how large a unit the farad is.

P16.10

Before: $Q = CV = 1.0 \text{ mC}$ and $U = \frac{1}{2}CV^2 = \frac{1}{2}(10 \times 10^{-6})(100)^2 = 0.050 \text{ J}$.

After. The charge is shared between two equal capacitors, which end at the same voltage. Each holds 0.50 mC, so

$$V = \frac{0.50 \text{ mC}}{10 \mu\text{F}} = 50 \text{ V}, \quad U = 2 \times \frac{1}{2}(10 \times 10^{-6})(50)^2 = 0.025 \text{ J}.$$

Missing energy. Exactly half the energy, 0.025 J, has disappeared. It was dissipated as heat in the resistance of the connecting wires, and partly radiated as electromagnetic waves while the charge surged across. Remarkably, the loss is always one-half, **whatever the resistance**. A smaller resistance gives a larger, shorter current pulse, and the total heat is the same. This is the same “half is lost” result as the conveyor belt of Problem P6.15.

P16.11

(a) Capacitance:

$$C = \frac{2U}{V^2} = \frac{2(200)}{(2000)^2} = 1.0 \times 10^{-4} \text{ F} = 100 \mu\text{F}.$$

(b) Charge: $Q = CV = (1.0 \times 10^{-4})(2000) = 0.20 \text{ C}$.

(c) Average current: $I = Q/t = 0.20/0.010 = 20 \text{ A}$.

P16.12

(a) Capacitance and charge.

$$A = 4\pi(15 \times 10^{-6})^2 = 2.83 \times 10^{-9} \text{ m}^2, \quad C = (0.010 \text{ F/m}^2)(2.83 \times 10^{-9}) = 2.8 \times 10^{-11} \text{ F} = 28 \text{ fF}$$

$$Q = CV = (2.83 \times 10^{-11})(0.070) = 2.0 \times 10^{-12} \text{ C}.$$

(b) Number of ions: $N = Q/e = 2.0 \times 10^{-12}/1.602 \times 10^{-19} = 1.2 \times 10^7$ ions.

(c) Potassium content of the cell.

$$V_{\text{cell}} = \frac{4}{3}\pi(15 \times 10^{-6})^3 = 1.41 \times 10^{-14} \text{ m}^3.$$

A concentration of 140 mM is 140 mol/m³, so

$$N_K = (140)(1.41 \times 10^{-14})(6.022 \times 10^{23}) = 1.2 \times 10^{12} \text{ ions.}$$

The fraction that must cross is $1.2 \times 10^7 / 1.2 \times 10^{12} = 10^{-5}$, about one in 100 000.

Significance. The membrane potential is set by a tiny charge imbalance confined to a thin layer at the membrane. The bulk fluids stay neutral, and their concentrations are essentially unchanged by charging the membrane. This is why physiologists can treat ion concentrations and membrane voltage almost independently, and why a single action potential barely changes the cell's ion content.

P16.13

Treat it as two capacitors in series: the air gap of thickness $d - t$ and the dielectric of thickness t , both of area A . They carry the same charge, and their voltages add:

$$\frac{1}{C} = \frac{d-t}{\epsilon_0 A} + \frac{t}{\kappa \epsilon_0 A} \Rightarrow C = \frac{\epsilon_0 A}{d-t+t/\kappa}. \checkmark$$

(Where the slab sits in the gap does not matter.)

Limits:

- $t = 0$: $C = \epsilon_0 A/d$, the empty capacitor. $\checkmark$
- $t = d$: $C = \kappa \epsilon_0 A/d$, completely filled. $\checkmark$
- $\kappa \rightarrow \infty$ (a metal slab): $C = \epsilon_0 A/(d - t)$. The metal simply shortens the gap, since there is no field inside a conductor. $\checkmark$

P16.14

Side by side: the two halves are in parallel, each with gap d and area $A/2$:

$$C_{\parallel} = \frac{\epsilon_0 (A/2)}{d} + \frac{\kappa \epsilon_0 (A/2)}{d} = \frac{\epsilon_0 A(1 + \kappa)}{2d}.$$

Layered (P16.13 with $t = d/2$):

$$C_{\text{series}} = \frac{\epsilon_0 A}{d/2 + d/(2\kappa)} = \frac{2\kappa \epsilon_0 A}{d(1 + \kappa)}.$$

Comparison:

$$\frac{C_{\parallel}}{C_{\text{series}}} = \frac{(1 + \kappa)^2}{4\kappa} \geq 1,$$

with equality only when $\kappa = 1$. For example, with $\kappa = 4$ the ratio is $25/16 = 1.56$. The **side-by-side** arrangement always gives more capacitance.

P16.15

(a) **Force from energy.** At constant Q , $U = \frac{Q^2}{2C} = \frac{Q^2 x}{2\epsilon_0 A}$, where x is the separation:

$$F = -\frac{dU}{dx} = -\frac{Q^2}{2\epsilon_0 A}.$$

The minus sign means the force tends to reduce x : the plates **attract** with magnitude $Q^2/(2\epsilon_0 A) = \frac{1}{2}QE$, where $E = Q/(\epsilon_0 A)$.

(b) **Why $\frac{1}{2}QE$.** The field $E = \sigma/\epsilon_0$ between the plates is produced by *both* plates, each contributing $\sigma/2\epsilon_0$. A plate cannot push or pull on itself, so each plate feels only the other plate's field, $E/2$. The force is therefore $Q(E/2)$.

(c) **MEMS actuator.** Using $F = \frac{1}{2}\epsilon_0 AV^2/d^2$:

$$F = \frac{(8.854 \times 10^{-12})(1.0 \times 10^{-6})(10)^2}{2(2.0 \times 10^{-6})^2} = 1.1 \times 10^{-4} \text{ N}.$$

That is about 0.1 mN, a large force for a device weighing micrograms. Electrostatic actuators move the micromirrors in digital light projectors and drive many MEMS devices.

P16.16

(a) **Field energy.** Outside the sphere, $E = kQ/r^2$, and inside it $E = 0$. Integrate over spherical shells of volume $4\pi r^2 dr$:

$$U = \int_R^\infty \frac{1}{2}\epsilon_0 \left(\frac{kQ}{r^2}\right)^2 4\pi r^2 dr = 2\pi\epsilon_0 k^2 Q^2 \int_R^\infty \frac{dr}{r^2} = \frac{2\pi\epsilon_0 k^2 Q^2}{R}.$$

Since $4\pi\epsilon_0 k = 1$, we have $2\pi\epsilon_0 k^2 = k/2$, so

$$U = \frac{kQ^2}{2R} = \frac{1}{2}Q \left(\frac{kQ}{R} \right) = \frac{1}{2}QV. \checkmark$$

The energy “stored in the capacitor” and the energy “stored in the field” are the same thing.

(b) **Classical electron.** Setting $\frac{ke^2}{2R} = m_e c^2$:

$$R = \frac{ke^2}{2m_e c^2} = \frac{2.82 \text{ fm}}{2} = 1.4 \text{ fm}.$$

This is the same order as the classical electron radius (the factor of 2 depends on how the charge is assumed to be distributed). Experiments show the electron is point-like down to below 10^{-18} m, far smaller. A point charge would have infinite field energy, a puzzle that classical physics cannot resolve. It is handled in quantum electrodynamics.

P16.17

(a) **Change in each capacitance.** $C \propto 1/d$, and the gaps become $1.99 \mu\text{m}$ and $2.01 \mu\text{m}$:

$$C_+ = 1.00 \times \frac{2.00}{1.99} = 1.00503 \text{ pF } (+5.0 \text{ fF}), \quad C_- = 1.00 \times \frac{2.00}{2.01} = 0.99502 \text{ pF } (-5.0 \text{ fF}).$$

(b) **Why measure the difference.** The difference, $C_+ - C_- \approx 10 \text{ fF} = 2C_0 \Delta d/d$:

- **doubles the signal** compared with a single capacitor;
- is **linear** in Δd to first order, since the second-order terms cancel;
- **cancels common-mode effects**, such as temperature changes, humidity and electronic drift, which change both capacitances in the same way.

(c) **Acceleration.** The suspension force must supply ma :

$$a = \frac{kx}{m} = \frac{(0.50)(10 \times 10^{-9})}{1.0 \times 10^{-9} \text{ kg}} = 5.0 \text{ m/s}^2 \approx 0.5 g.$$

P16.18

(a) **Capacitance.** The filled and empty parts of the capacitor are in parallel:

$$C(x) = \frac{\kappa \epsilon_0 w x}{d} + \frac{\epsilon_0 w (L - x)}{d} = \frac{\epsilon_0 w}{d} [L + (\kappa - 1)x].$$

(b) **Force.** If the slab moves in by dx at constant V :

- the capacitance rises by dC , and the battery supplies work $V dQ = V^2 dC$;
- the stored energy rises by $dU = \frac{1}{2} V^2 dC$.

The remaining energy, $\frac{1}{2} V^2 dC$, is the mechanical work $F dx$ done on the slab:

$$F = \frac{1}{2} V^2 \frac{dC}{dx} = \frac{(\kappa - 1) \epsilon_0 w V^2}{2d}.$$

This is independent of x and directed inward. It arises physically from the fringing field at the edge of the plates acting on the slab's polarised charges.

(c) **Numbers:**

$$F = \frac{(2.0)(8.854 \times 10^{-12})(0.10)(1000)^2}{2(1.0 \times 10^{-3})} = 8.9 \times 10^{-4} \text{ N}.$$

This is a small force. The same principle, with liquid dielectrics, is used in electrowetting displays and in “lab-on-a-chip” devices that move droplets around.

All worked examples and practice problems in these notes are original. Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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